

# Maksuda Aktar Toma

June 3, 2026

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**Google Scholar:** <https://scholar.google.com/citations?user=ukmoEDgAAAAJ>

## Research interests

Change Point Detection, AI in Agriculture, Deep Learning, Machine Learning, Data Visualization, Causal Inference

## Education

**University of Nebraska – Lincoln** 2025 – Present

PhD in Statistics (in progress) CGPA: 3.732/4.00  
Advisors: Dr. Reka Howard & Dr. Sruti Das Choudhury

**University of Nebraska – Lincoln** 2023 – 2025  
Master of Science in Statistics CGPA: 3.634/4.00

**University of Dhaka** 2020 – 2021  
Master of Science in Statistics CGPA: 3.221/4.00

**University of Dhaka** 2016 – 2020  
Bachelor of Science in Statistics CGPA: 3.284/4.00

## Research experience

### Hyperspectral Maize Tassel Detection and Phenotyping Using YOLOv8 and Vision-Language Models

Mentor: Dr. Reka Howard and Dr. Sruti Das Choudhury (University of Nebraska – Lincoln) Jan 2026 – Present

Developing a deep learning pipeline for automated tassel detection and phenotype extraction in maize plants using hyperspectral imaging data captured across multiple plants and growth stages. Performing spectral band analysis across 243 bands to identify optimal detection wavelengths, and exporting calibrated single-band grayscale images for model input. Annotating images using Roboflow and training a YOLOv8 object detection model on the UNL HCC Swan GPU cluster. Extending the pipeline with Vision-Language Models (VLMs) to enable semantic description and interpretation of tassel development, bridging detection outputs with agronomic insight. [here](#)

### Feeding Behavior-Based Detection of Pig Health Status Using Machine Learning Modeling

Mentor: Dr. Sruti Das Choudhury and Dr. Tami Brown-Brandl (University of Nebraska – Lincoln) Aug 2025 – May 2026

Applied segmented regression and machine learning to model longitudinal feeding behavior patterns from a USDA swine production dataset. Identified behavioral change points associated with disease onset and recovery, and developed multiclass health-status classification models using LASSO, Ridge, and Random Forest methods. Conducted cross-validation, hyperparameter tuning, and diagnostic evaluation. Manuscript under review at *Biosystems Engineering*. [here](#).

### **100 Years of Pies vs. Bars**

Mentor: Dr. Susan Vanderplas (University of Nebraska – Lincoln) Jan 2025 – Aug 2025

Investigated 100 years of pie vs. bar chart research on graphical perception. Analyzed conditions under which each chart type optimizes data interpretation, contributing to evidence-based visualization guidance. [here](#).

## Statistical Consulting

**Statistical Consultant, Department of Statistics (University of Nebraska – Lincoln)** Fall 2026 (Upcoming)

SC3L Help Desk (Statistical Consulting, Collaboration, and Communication Lab)

Provide statistical consulting services to graduate students, faculty, and researchers across university departments. Advise on study design, data analysis, and interpretation of results, applying methods from regression, mixed models, and applied statistics to diverse research problems.

## Teaching experience

**Graduate Teaching Assistant, Department of Statistics (University of Nebraska – Lincoln)** Fall 2023 – Summer 2026

STAT 218: Introduction to Statistics

STAT 380: Statistics and Applications

Instructor for introductory and intermediate undergraduate statistics courses covering probability concepts, statistical inference, and regression analysis. Developed, administered, and graded assignments and exams, providing detailed feedback to strengthen student understanding.

**Learning Assistant, College of Agricultural Sciences and Natural Resources (CASNR) (University of Nebraska – Lincoln)** Fall 2025 – Present

SCIL 101: Science and Decision Making - For a Complex World

Mentoring students across several project groups, guiding them through experiential learning and a semester-long showcase project. Leading active learning discussions, assessing student learning, and providing feedback on assignments and final projects.

## Publications

### **Feeding Behavior-Based Detection of Pig Health Status Using Machine Learning Modeling (Under Review)**

Maksuda Aktar Toma, Tami Brown-Brandl, Gary Rohrer, Sruti Das Choudhury  
*Biosystems Engineering*, 2026.

### **Factors influencing non-academic achievements of Bangladeshi students - An online questionnaire-based study (Preprint)**

Md. Rabiul, Md. Hasibur Rahman, Maksuda Aktar Toma, Pranta Das  
*Research Square*, 2021. DOI: [10.21203/rs.3.rs-410311/v1](https://doi.org/10.21203/rs.3.rs-410311/v1)

## Awards & Honors

### **Student Research Days Winner, Office of Research and Innovation April 2026**

University of Nebraska – Lincoln

Winner, Spring 2026 Student Research Days competition; rated by 40+ faculty and postdoc reviewers on research scholarship and presentation skills. Includes \$400 travel grant. [[UNL Newsroom](#)]

### **Second Place, Student Award for Oral Presentations (\$400) May 2026**

Workshop on Applied Statistics in Agriculture & Natural Resources, University of Nebraska – Lincoln

## Talks

### **Change-Point Detection in Feeding Behavior Transitions for Multi-Stage Pig Health Classification Using Machine Learning**

*Oral Presentation (2nd Place, Student Award)*, Workshop on Applied Statistics in Agriculture & Natural Resources, University of Nebraska – Lincoln May 2026

Session: AI in Agriculture

### **Feeding Behavior-Based Detection of Health Status in Pigs Using Machine Learning Modeling**

*Oral Presentation*, 2026 AI in Agriculture Conference, North Carolina State University March 2026

Session: AI for Livestock Monitoring

### **100 Years of Pies vs. Bars**

2025 International Indian Statistical Association

June 2025

Kansas Math Symposium 2025

April 2025

## Industry experience

### **Association for Social Advancement (ASA)**

Dhaka, Bangladesh

Research Officer

Apr 2023 – Jul 2023

Conducted mixed-method analyses to evaluate program outcomes and risks, designed surveys, ensured data quality, and presented actionable insights and reports to stakeholders

**Centre for Governance Studies (CGS)**

Dhaka, Bangladesh

Research Assistant

Apr 2022 – Mar 2023

Conducted qualitative research on good governance, led focus groups for a U.S. State Department project on misinformation, and co-authored articles on societal issues

**Ethical Trading Initiative (ETI)**

Dhaka, Bangladesh

Intern

Jan 2021 – Jul 2021

Performed data analysis, data validation, reporting, and qualitative research (FGD/KII).

**Programming and Tools**

Skills

Proficient in: R, Python, MATLAB, TensorFlow, SAS, STATA, SPSS, SQL, PowerBI

Familiar with: C, C++, Fortran

**Languages**

English, Bengali, Hindi

**Treasurer, Statistics Graduate Student Association (GSA), UNL**

July

Service and outreach

2024 – Present

Manage the budget, coordinate funding requests, and support academic and social events in collaboration with the executive board.

**General Secretary, Dhaka University Debating Society (DUDS)**

Oct 2021 – Dec 2022

Organized multiple national and international debate festivals, managing logistics, partnerships, and teams. Won multiple championships in national and international tournaments.

**American Statistical Association (ASA)**

July 2024 – Present

Professional

memberships

Attended AMSTAT Nebraska Travelling Course "From Big Data to Better Insights: A Primer on Machine Learning Methods in a Data-Centric World."

**International Indian Statistical Association (IISA)**

Jun 2025 – Present

Attended the conference and took the short courses "Multimodal Causal Inference for Data Science and Biomedical Research" and "Boosting R Code performance via C++ Integration within Rstudio".